

Glutathione

A tripeptide made in the liver from three amino acids: glutamine, cysteine and glycine forming an active antioxidant known as reduced glutathione (GSH) and an inactive form known as oxidized glutathione (GSSG). The key functions include acting as a powerful antioxidant by neutralizing free radicals and the prevention of oxidative stress, liver detoxification (phase II conjugation), binds to toxins, heavy metals and drugs, making them water soluble for excretion, cellular protection by binding to the mitochondria and DNA from damage, supports proper function of T-cells and other immune cells and maintains a balanced internal oxidative-reductive state.

Contraindications of Glutathione

Known Allergy or Hypersensitivity:

Individuals who have had allergic reactions to glutathione, sulfa, or any of its components should not use it. If there are any signs of an allergic reaction, the patient should be directed to the nearest emergency room immediately.

Asthma:

Glutathione inhalation can exacerbate asthma symptoms and may trigger bronchospasm. It is contraindicated in people with asthma when used via inhalation.

Pregnancy and Breastfeeding:

There is insufficient safety data on the use of glutathione during pregnancy and lactation. It should be avoided unless clearly needed and prescribed by a healthcare provider.

Chemotherapy:

Some evidence suggests that glutathione may interfere with certain chemotherapy agents, particularly those that rely on oxidative stress to kill cancer cells. Its use should be carefully evaluated in cancer patients undergoing treatment.

Organ Transplant Patients:

Due to its immune-modulating effects, glutathione may theoretically interfere with immunosuppressive therapy. Its use is generally not recommended in transplant patients unless under strict medical supervision.

Administration

Intramuscular (IM) Injection

- Needle Length:
 - 1 to 1.5 inches (25–38 mm), depending on body fat
- Needle Gauge:
 - 22 to 25 gauge
- Common Sites:
 - Deltoid (upper arm), gluteus (buttock), or vastus lateralis (thigh)

Intravenous (IV) Injection or IV Push

- Needle/Catheter Gauge:
 - 20 to 25 gauge for a butterfly or IV catheter
 - Use a smaller gauge (higher number) for slower push; larger gauge (lower number) for faster flow

- Site:
 - Typically a vein in the arm or hand
- Method:
 - Often given slowly over 5–15 minutes via syringe (IV push), or as part of an IV drip (e.g., with vitamin C)

Dosage

1. Intravenous (IV) Glutathione Dosage

Use	Typical Dosage	Frequency
General antioxidant support	600–1,200 mg	1–2 times per week
Skin lightening (off-label use)	1,000–2,400 mg	1–3 times per week for several weeks/months
Chronic illness (e.g., Parkinson's, liver disease, heavy metal detox)	600–2,500 mg	2–3 times per week under supervision

⚠ Often combined with **vitamin C** (ascorbic acid) to support recycling and stability.

2. Intramuscular (IM) Glutathione Dosage

- **Typical range:** 200–1,200 mg per injection *start low and increase if needed every 4 weeks
 - **Frequency:** 1–3 times per week
 - IM injections are slower-absorbing than IV but still effective for general antioxidant support or maintenance.
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Patient Education

Injection Site Reactions

- Mild pain, redness, swelling, or bruising at the injection site may occur.
- Use a warm compress if discomfort occurs. Report any signs of **severe swelling**, **persistent pain**, or **pus**, which could indicate infection or abscess formation.

Allergic Reactions

- Although rare, allergic reactions are possible.
 - **Seek immediate medical attention** if you experience:
 - Rash or hives
 - Itching or swelling (especially of the face/tongue/throat)
 - Dizziness or lightheadedness
 - Difficulty breathing
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Risk of Hematoma

- If you have a bleeding disorder or take anticoagulants (e.g., warfarin, aspirin), inform your provider.
- There is an increased risk of bruising or bleeding at the injection site.

Dizziness or Lightheadedness

- Some individuals may feel **faint** or **dizzy** immediately after injection.
- Remain seated or lying down for 10–15 minutes post-injection to avoid falls.

Use Caution if Pregnant or Breastfeeding

- Safety of glutathione injections during pregnancy or lactation has not been firmly established.
- Always consult your healthcare provider before use.

Potential for Drug Interactions

- Inform your provider of all medications and supplements you are taking.
- Caution is advised if using other antioxidants or chemotherapy agents.

Overuse or High-Frequency Injections

- Excessive use may disrupt the body's natural antioxidant balance.
- Stick to the prescribed dosing schedule.
- Glutathione is not FDA-approved for anti-aging, skin lightening, or athletic performance enhancement — use only under medical supervision.



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Hydration and Diet

- Stay well-hydrated before and after injections to support detoxification and renal clearance.
- A healthy liver and kidney function are important for proper glutathione metabolism.

When to Call Your Provider

Contact your provider immediately if you experience:

- Signs of **infection** at the injection site
- **Yellowing of the skin or eyes** (possible liver issues)
- **Persistent nausea, fatigue, or dark urine**
- Any symptoms that concern you after receiving the injection



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Glutathione IV Infusion Combinations

Enhanced Hangover					
Medication	Strength (mg/mL)	Osmolarity MDV (with Preservative) mOsm/mL	Dose Volume (cc)	mOsms	Total Dose (mg)
0.9% Saline	-	0.31	1000	310	-
B-Complex	100	2.04	1	2.04	100
Methylcobalamin	5	2	1	2	5
Magnesium Chloride	200	2.95	1	2.95	200
Toradol	30	-	1	PUSH	30
Ondansetron	2	-	2	PUSH	4
Famotidine	10		2	PUSH	20
		Total Volume	1003		
		Total mOsms		317	
		Final mOsms		316	

Enhanced Myers					
Medication	Strength (mg/mL)	Osmolarity MDV (with Preservative) mOsm/mL	Dose Volume (cc)	mOsms	Total Dose (mg)
0.9% Saline	-	0.31	1000	310	-
B-Complex	100	2.04	1	2.04	100
Vitamin C	500	6.05	2	12.1	1000
Magnesium	200	2.95	1	2.95	200
Glutathione		-	5	PUSH	1000
		Total Volume	1009		
		Total mOsms		327	
		Final mOsms		324	



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Enhanced Immunity					
Medication	Strength (mg/mL)	Osmolarity MDV (with Preservative) mOsm/mL	Dose Volume (cc)	mOsms	Total Dose (mg)
0.9% Saline	-	0.31	1000	310	-
B-Complex	100	2.04	1	2.04	100
Vitamin C	500	6.05	2	12.1	1000
Zinc	0.5	0.011	5	0.055	2.5
Glutathione	200	2.16	5	PUSH	1000
Toradol	30	-	1	PUSH	30
Ondansetron	2	-	2	PUSH	4
Famotidine	10		2	PUSH	20
		Total Volume	1013		
		Total mOsms		324	
		Final mOsms		320	

Beauty					
Medication	Strength (mg/mL)	Osmolarity MDV (with Preservative) mOsm/mL	Volume (cc)	mOsms	Total Dose (mg)
0.9% Saline	-	0.31	1000	310	-
Biotin	10	0.22	1	0.22	10
Vitamin C	500	6.05	2	12.1	1000
Zinc	0.5	0.011	5	0.055	2.5
Glutathione	200	2.16	5	PUSH	1000
		Total Volume	1013		
		Total mOsms		322.4	
		Final mOsms		318.2	



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Enhanced NAD					
Medication	Strength (mg/mL)	Osmolarity MDV (with Preservative) mOsm/mL	Volume (cc)	mOsms	Total Dose (mg)
0.9% Saline	-	0.31	1000	310	-
NAD+	100	-	5	-	500
B-Complex	100	2.04	1	2.04	100
Methylcobalamin	5	2	1	2	5
Vitamin C	500	6.05	4	24.2	2000
Glutathione	200	2.16	5	PUSH	1000
Taurine	50	0.95	6	PUSH	300

Total Volume	1011	
Total mOsms		338
Final mOsms		335